

Functions & Graphs

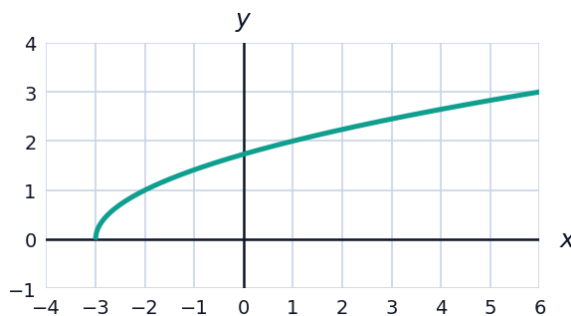


Evaluating functions

- 1 If $f(x) = 3x - 2$, find $f(4)$.
- 2 If $f(x) = x^2 + 2x$, find $f(-3)$.
- 3 Consider $f(x) = 2x^2 - 5x + 1$. Evaluate:
a $f(0)$ **b** $f(2)$

Domain and range

- 4 The graph of $y = \sqrt{x + 3}$ is shown below. State the domain and range based on the graph, then verify the domain algebraically.



- 5 Find the domain of $f(x) = \frac{1}{x - 5}$.
- 6 Find the domain of $f(x) = \sqrt{2x - 8}$.

Composition of functions

- 7 If $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $(f \circ g)(2)$.
- 8 If $f(x) = x + 5$ and $g(x) = 3x$, find $(g \circ f)(x)$ as a simplified expression.
- 9 If $f(x) = x^2$ and $g(x) = x - 4$, find $(f \circ g)(x)$ and $(g \circ f)(x)$, and state whether they are equal.

Inverse functions

- 10 Find the inverse of $f(x) = 4x - 8$.
- 11 Find the inverse of $f(x) = \frac{x-3}{2}$.
- 12 Verify that $f(x) = 5x + 1$ and $g(x) = \frac{x-1}{5}$ are inverses of each other by computing $f(g(x))$.
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Transformations of graphs

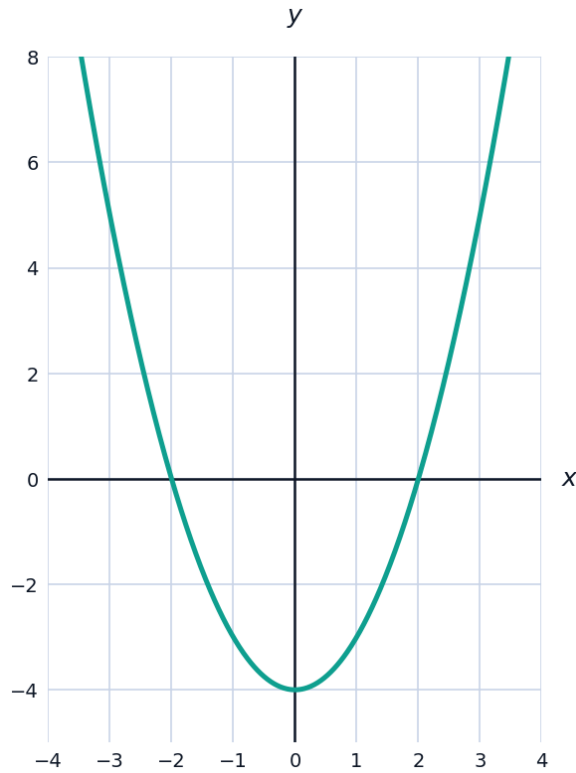
- 13 The graph of $y = f(x)$ is shifted 4 units left and 2 units down. Write the new function in terms of f .
- 14 If $f(x) = x^2$ and $g(x) = -f(x) + 3$, describe the transformation from f to g .
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Piecewise functions

- 15 A function is defined by $f(x) = x + 2$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$. Find $f(-3)$ and $f(3)$.
- 16 A function is defined by $f(x) = 2x$ for $x \leq 2$ and $f(x) = x + 5$ for $x > 2$. Find $f(2)$ and $f(5)$, and state whether f is continuous at $x = 2$.
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Reading function graphs

- 17 The graph of $f(x) = x^2 - 4$ is shown below. Use the graph to estimate the x -intercepts and y -intercept, then find them exactly.



Even, odd, and other function properties

- 18 Determine whether $f(x) = x^2 - 3$ is even, odd, or neither, using $f(-x)$.
- 19 Determine whether $f(x) = x^3 - x$ is even, odd, or neither.

A well-designed word problem: modeling a car rental cost

- 20 This question has three parts. A rental car company models the cost of a rental as $C(d) = 45d + 20$ dollars, where d is the number of days rented. (a) Find $C(5)$, and interpret its meaning in context. (b) Define $P(d) = C(d) - 200$, representing profit relative to a fixed cost of 200 dollars per rental period. Write $P(d)$ explicitly and find $P(5)$. (c) Find the inverse function $C^{-1}(x)$, and use it to determine how many days someone rented the car if they were charged 335 dollars.